

Typical Classification Dataset

Balanced Dataset

Support of each unique label category is equal.

Unbalanced Dataset | Imbalance Dataset.

Support of each unique label is different.

CLASS IMBALANCE EXISTS IN SAMPLE.

Evaluation metrics in Supervised Learning

} Check the quality of the model based on agreed criteria

Regression

- adjusted R^2
- fn name in sklearn is `r2_score`
- default metric for model score f^2 .

Classification

- Accuracy
- Precision
- Recall
- F1-score

when to use which evaluation metric for classification

Balanced Dataset

Accuracy →

- for generalization check
- CL check with entire dataset.

Imbalance Dataset

Guideline by PN

Statistical Approach

[applicable for both binary & multi-class classification]

Accuracy → filter out generalized model based on CL criteria

F1 Score → for CL check with entire dataset.

Domain Approach

[applicable for Binary classification]

Accuracy → filter out generalized model based on CL criteria

PR pair → for CL check with entire dataset.

Confusion matrix (CM)

Error matrix in tabular form to visualize the performance of the classification model.

eg: automobile industry

$$S(\text{approve}) = 95 + 2 = 97$$

$$S(\text{reject}) = 37 + 66 = 103$$

$$\Sigma S = 200$$

$$SL = 0.2$$

$$\text{correct predictions} = 95 + 66 = 161$$

to visualize the performance of

actual values
ground truth

		predicted values	
		a	reject
actual values ground truth	approve	95	2
	r	37	66

$$\text{misclassified records} = 37 + 2 = 39$$

Accuracy: applicable for entire model.
(go diagonal) FOCUS ON OVERALL CORRECTNESS OF THE MODEL

$$acc_{(model)} = \frac{\sum \text{diagonal values}}{\sum \text{all values}}$$

Precision: Precision FOCUS on BETTER
(go vertical) PREDICTION QUALITY in output.
applicable for each class label.

iris dataset

P_{setosa}

$P_{versicolor}$

$P_{virginica}$

Recall: applicable for each class label.
(go horizontal)

Recall FOCUS on FACTUAL CONSISTENCY
in outputs.

F1 Score: Its the HARMONIC mean of precision
and recall.

$$f1_{score} \text{ (eachlabel)} = \frac{2 * P_{eachlabel} * R_{eachlabel}}{P_{eachlabel} + R_{eachlabel}}$$

Email Inbox Classification

Primary

Social

Promotions

Updates

$$s(\text{Primary}) = 102$$

$$s(\text{Social}) = 105$$

$$s(\text{Promotions}) = 208$$

$$s(\text{Updates}) = 400$$

$$CL = \cancel{0.6}$$

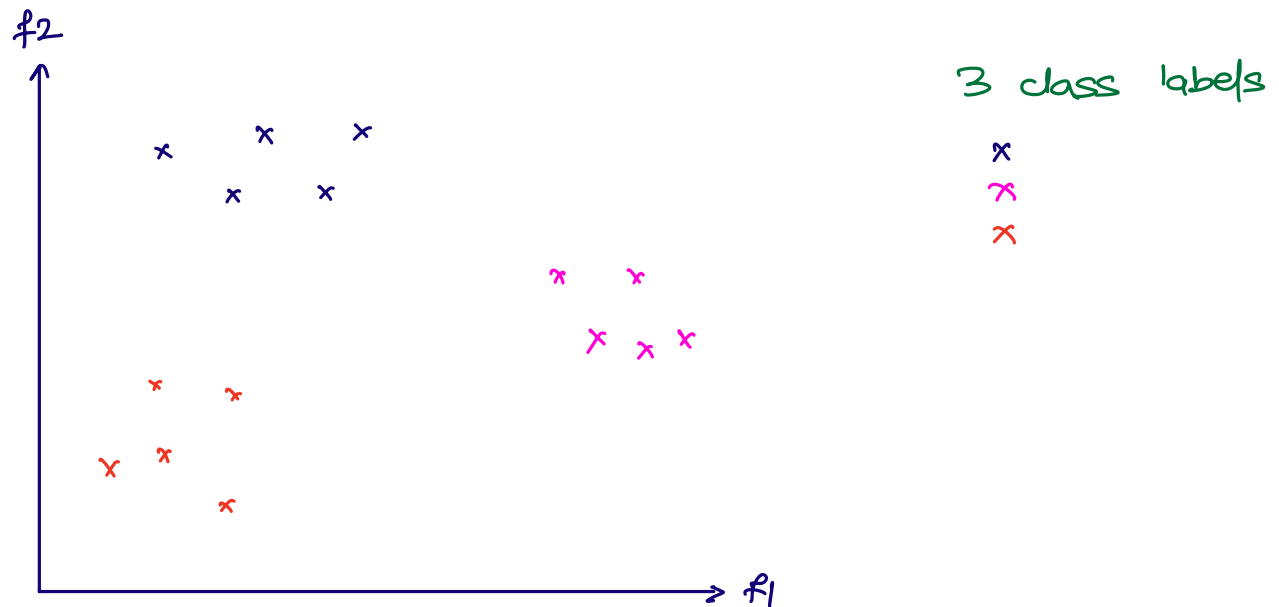
$$SL = \underline{\underline{0.4}}$$

100	2	0	0
0	7	200	1
0	100	3	2
0	0	0	400

$$\text{avg F1Score} = \geq 0.6 \} \rightarrow \text{Yes} \rightarrow \text{Approve}$$

KNN (K-Nearest Neighborhood)

→ Supervised Learning algo. —→ Regression
Classification



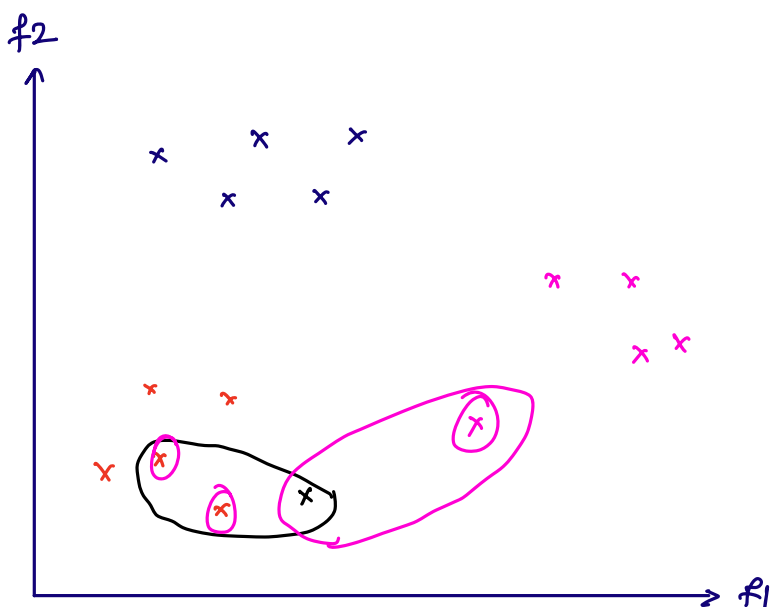
→ Instance - Based model | Instance-based algorithm.

model doesn't extract / discover pattern.

It simply memorize training data.

∴ model size = training data size.

→ How KNN performs prediction?



Euclidean Distance Formula

$$\sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$$

- ① Calc the distance betⁿ unknown pt and all known data points
- ② Sort the data points based on distance in ascending order
- ③ Select first 'k' points (k=3)
- ④ Perform election
 - Blue 'x' → 0
 - Red 'x' → 2 —→ *
 - Magenta 'x' → 1

⑤ Whichever wins, the unknown pt belongs to that class.

Prediction Loop

